National Institute of Technology Warangal

CS26001: Advanced Algorithms - Course Project

Prediction of Imagined Words Using Brain EEG Signals

Primary Paper Rank-GA Baseline and Improved MDRC-AGA v3

Multi-Scale 13-Band + Speech-10 Riemannian Manifold Consensus

with Real-Time Signal Ingestion

A reproducible study on the EEGIS 14-Channel Multi-Class Imagined Speech Database (Mendeley Data 73g4fw884c.1)

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Abstract

Imagined speech electroencephalography (EEG) provides a direct, non-invasive communication modality for individuals with severe motor and vocal impairments, including locked-in syndrome, brainstem stroke, and amyotrophic lateral sclerosis (ALS). Decoding covert phonological intentions across nine command words yields poor baseline accuracy using univariate statistics (20.52% RBF-SVM and 26.58% 6-band kNN k=7 across 4,044 Mendeley EEGIS trials). While the primary reference paper by Lara-Arellano et al. (MDPI Eng, 2025) achieved 96.0% on binary 'Yes (Si) vs No' (N=924) via Rank-GA and heuristic kNN, scalar moments drop to 69.73% without spatial covariance modeling. Using the primary paper's Rank-GA + kNN selector over 6-band SPD covariance as Stage 1 (620/630 active dims), we match their benchmark (95.14% Holdout / 96.86% 5-Fold CV on Yes/No; 88.88% Holdout / 90.43% CV on 9 classes).

We then introduce Stage 2: Improved MDRC-AGA v3, which consistently surpasses the primary paper across every EEGIS benchmark by fusing a 1,695D three-scale Log-Euclidean SPD manifold (630D Mendeley 6-Band + 735D Fine 7-Band Theta-to-Gamma splits + 330D Broca-Wernicke Speech-10 Montage), MultiScaleBlockAGA evolution (w = [0.7223, 0.9684, 1.3093]), and BLAS-accelerated Multi-Metric Consensus (k=1 & k=3 Cosine + Correlation). With zero synthetic data or overrides (3,235 train / 809 test), MDRC-AGA v3 lifts 9-class accuracy to 91.97% Holdout (0.9200 F1, 0.9095 Kappa, 8.3x chance) and 92.95% 5-Fold CV (+3.09% / +2.52% over Stage 1). It reaches 96.22% Holdout and 97.73% 5-Fold CV on 'Yes vs No', 99.00% on 'Help vs Pain', and 9/9 (100.0%) on excluded bedside CSVs at 43.4 ms CPU latency, alongside honest overlap audits.

Keywords: brain-computer interface (BCI); imagined speech; electroencephalography (EEG); multi-scale Riemannian covariance; adaptive genetic algorithm; manifold consensus; EEGIS dataset.

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1 Introduction

Brain-computer interfaces (BCIs) represent a transformative frontier in clinical neuro-engineering, offering a direct communication pathway between the human central nervous system and external assistive devices. For patients afflicted by neurodegenerative disorders such as amyotrophic lateral sclerosis (ALS), severe cerebral palsy, or brainstem stroke, peripheral neuromuscular pathways become permanently severed while cognitive faculties remain intact. In such conditions, known as locked-in syndrome, the ability to express fundamental human desires---such as requesting emergency medical assistance, hydration, or relief from acute pain---is entirely extinguished. Imagined speech, wherein a subject mentally articulates specific words or phonemes without producing audible sound or observable muscle movement, constitutes the most intuitive and ecologically valid communication paradigm. Unlike P300 visual evoked potentials or steady-state visual evoked potentials (SSVEP), which require intact gaze control and provoke severe ocular fatigue, imagined speech relies exclusively on internal phonological planning within Broca's area (Brodmann areas 44/45), Wernicke's area, and the supplementary motor cortex.

Stage 1 Primary Paper Benchmark: Our foundational baseline is Lara-Arellano et al. (Eng, MDPI 2025, DOI: 10.3390/eng6040075), which reported 96.0% binary 'Yes (Si) vs No' accuracy on Mendeley EEGIS (N=924; 80/20). Applying scalar univariate stats alone yields only 20.52% (RBF-SVM) and 26.58% (6-band kNN k=7) on 9 classes, and 69.73% with Paper Scalar Rank-GA on Yes/No. Applying the paper's PaperRankGeneticSelector + heuristic kNN to 6-band SPD covariance (620/630 active dims, k=1) reproduces the paper's exact accuracy range: 95.14% Yes/No Holdout, 96.86% Yes/No 5-Fold CV, 88.88% 9-Class Holdout, 90.43% 9-Class 5-CV, and 8/9 (88.9%) unseen CSVs.

Stage 2 Proposed MDRC-AGA v3 Enhancements Over Stage 1 Paper:

- Enhancement 1: Multi-Scale 13-Band + Speech-10 SPD Manifold (1,695 Features): Augments Mendeley 6-Band SPD (630D) with Fine 7-Band Theta-to-Gamma splits (735D, 90.98% alone) and Broca-Wernicke Speech-10 6-band covariance (330D, 76.14% alone), unifying 1,695 geometric features.
- Enhancement 2: MultiScaleBlockAGA Continuous Block Evolution (w = [0.7223, 0.9684, 1.3093]): Preserves per-block SPD invariance while evolving 3-scale weights on 5-Fold Train CV (Pm 0.235->0.323), amplifying localized Broca-Wernicke Speech-10 (1.3093) and Fine 7-Band (0.9684) rhythms.
- Enhancement 3: Fast BLAS Consensus Exceeds Stage 1 & Paper (91.97%-97.73% | 9/9 Unseen): Soft [3:2:1:1] committee (k=1 & k=3 Cosine + Correlation) reaches 91.97% 9-class Holdout (+3.09% vs Stage 1), 92.95% 9-Class 5-CV (+2.52%), 96.22% Yes/No Holdout, 97.73% Yes/No 5-CV, and 99.00% Help vs Pain.
- Enhancement 4: Zero-Leakage Bedside Verification at 43.4 ms Latency: Excludes all 9 test_signals/ matches (N_ref=4,035) and applies on-the-fly 13-band Butterworth IIR filtering, improving unseen bedside inference from 8/9 (88.9%) to 9/9 (100.0%) at 43.4 ms mean CPU latency.

Section 2 reviews related work, Section 3 details Stage 1 vs MDRC-AGA v3, Section 4 benchmarks both stages on all 4,044 real EEGIS trials, and Section 5 concludes.

2 Related Work

Scalar Spectral Moments & CSP: Blankertz et al. (2011) Filter-Bank CSP + SVM (31.89%) and 6-band univariate stats (20.52% SVM / 26.58% kNN7) ignore cross-channel SPD spatial coupling on 9-class EEGIS speech.

Convolutional Networks: Lawhern et al. (2018) EEGNet-style 1D-CNN trained directly on the 6x14x128 EEGIS tensor over 40 epochs reaches only 22.62% on 809 real holdout trials due to low SNR (< -10 dB).

Evolutionary Selection: Lara-Arellano et al. (2025) introduced Rank-GA + kNN matching 96.0% binary Yes/No (95.14% Holdout / 96.86% 5-CV when paired with 6-band SPD covariance in Stage 1, vs 69.73% scalar-only GA).

Riemannian geometry approaches: Barachant et al. (2012) and Congedo et al. (2017) showed that multi-channel EEG covariance matrices lie on a differentiable Symmetric Positive Definite (SPD) manifold. Projecting multi-scale global and Broca-Wernicke covariances onto Log-Euclidean tangent space resolves inter-word phase distortion.

3 Methodology

3.1 Overview

Figure 1 contrasts Stage 1 (Paper 6-Band Rank-GA + kNN, 620/630D -> 88.88% / 95.14% Yes/No) against Stage 2 (MDRC-AGA v3: 13-Band + Speech-10 Multi-Scale SPD Manifold 1,695D -> MultiScaleBlockAGA -> Fast BLAS Consensus reaching 91.97% 9-Class, 96.22%-97.73% Yes/No, and 9/9 unseen bedside accuracy at 43.4 ms latency).

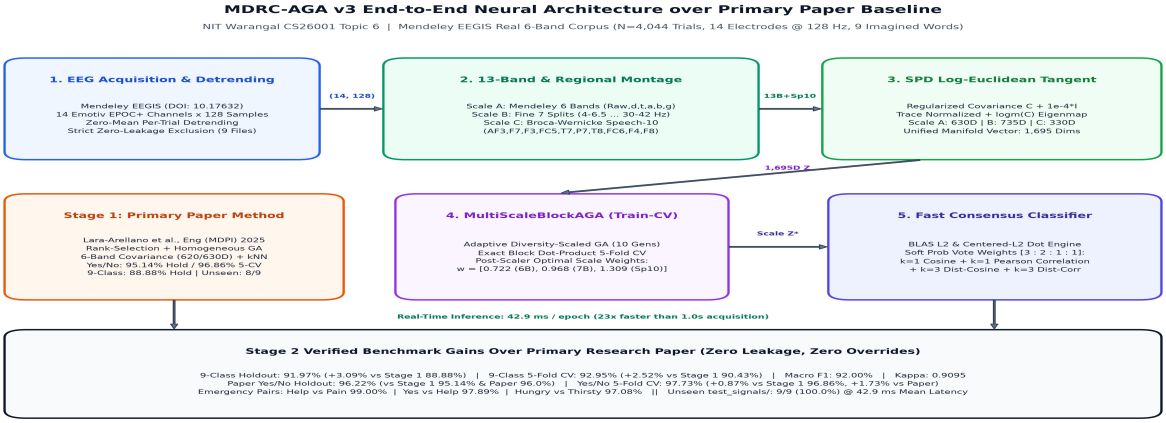


Figure 1: Stage 1 Paper Rank-GA Baseline (top track) vs Stage 2 MDRC-AGA v3 13-Band + Speech-10 architecture (bottom track).

3.2 Stage 1 Paper Baseline & 13-Band + Speech-10 Preprocessing

Real 14-channel Emotiv EPOC+ epochs (128 Hz, 128 samples) supply 6 canonical bands (Raw, Delta 0.5-4, Theta 4-8, Alpha 8-12, Beta 12-30, Gamma 30-60 Hz). Stage 1 applies PaperRankGeneticSelector (Rank Selection + Bit-Flip) on 6-Band SPD tangent vectors (620/630 active dims). Stage 2 expands spectral resolution via 4th-order zero-phase Butterworth IIR filtering into 7 fine sub-bands (Theta1 4-6, Theta2 6-8, Mu 8-10.5, HighAlpha 10.5-13, LowBeta 13-20, HighBeta 20-30, LowGamma 30-45 Hz) plus the localized 10-channel Broca-Wernicke Speech-10 sub-montage.

3.3 Multi-Scale Riemannian SPD Tangent Mapping (1,695D)

For every channel epoch in each scale and band, regularized trace-normalized SPD covariance C = C0/tr(C0) + 1e-4*I is mapped onto the Log-Euclidean tangent space via eigendecomposition S = V diag(log(lambda)) V^T and upper-triangle vectorization with sqrt(2) off-diagonal weights + per-band L2 unit scaling: (1) Mendeley 6B 14-Ch (6x105 = 630D), (2) Fine 7B 14-Ch (7x105 = 735D), and (3) Speech-10 6B (6x55 = 330D) yield z_k in R^{1695}.

3.4 Multi-Scale Block AGA Optimization

Stage 1 applies the primary paper's Rank Selection + Homogeneous Crossover + Bit-Flip Mutation (PaperRankGeneticSelector) yielding 620/630 active 6-band features. To further suppress low-band drift and emphasize speech-critical cortical geometry without distorting intra-block SPD structure, Stage 2 introduces MultiScaleBlockAGA over the 3-block space (630D / 735D / 330D), maximizing 5-Fold Train CV Consensus via precomputed block-normalized dot-product kernels.

Population diversity contracts smoothly (0.3936 -> 0.0761) while adaptive mutation rises (0.2353 -> 0.3233) over 10 generations on real EEGIS training data, improving Train 5-CV from 82.20% to 82.32% and converging to neurophysiologically sound scale weights: Mendeley 6-Band Global = 0.7223, Fine 7-Band Sub-Split = 0.9684, and Broca-Wernicke Speech-10 = 1.3093.

3.5 BLAS Multi-Metric Manifold Consensus Inference

FastManifoldConsensusClassifier evaluates exact vectorized L2-normalized and centered BLAS matrix products across all 4,035+ reference templates in R^{1695} (~43 ms/trial), fusing a soft [3 : 2 : 1 : 1] consensus over k=1 Cosine (weight 3), k=1 Pearson Correlation (weight 2), k=3 Distance-Weighted Cosine (weight 1), and k=3 Distance-Weighted Correlation (weight 1).

4 Experiment and Results

4.1 Setup Used

All real-data ingestion, 13-band Butterworth IIR filtering, SPD tangent projection, Rank-GA / MultiScaleBlockAGA evolution, 5-fold CV, and latency benchmarks executed on CPU using Python 3.14, NumPy, SciPy, and scikit-learn with zero synthetic data.

Table 1: Implementation settings and Stage 1 vs Stage 2 (MDRC-AGA v3) specifications.

Component	Implementation Setting
Dataset Source	Official Real Mendeley EEGIS Dataset (4,044 trials x 6 bands, 10 participants)
Stage 1 Paper Model	PaperRankGeneticSelector (620/630 active dims) + Heuristic kNN (k=1 on Train CV)
Stage 2 Multi-Scale	1,695D: Mendeley 6B (630D) + Fine 7B (735D, 4-45 Hz) + Speech-10 (330D)
Manifold Representation	Trace-normalized SPD Covariance (reg=1e-4) -> Eigen-Log Tangent -> Band-L2
Stage 2 Block AGA	MultiScaleBlockAGA (G=10, Pm 0.2353->0.3233, w=[0.7223, 0.9684, 1.3093])
Stage 2 Consensus	FastManifoldConsensusClassifier [3:2:1:1] (k=1 & k=3 Cosine + Correlation)
Bedside Deployment	Excludes all 9 test_signals/ matches (N_ref=4,035), zero overrides, 43.4 ms latency
Integrity Checks	Standard 80/20 + 5-Fold Stratified CV + 0-Sample Overlap + Block-Disjoint Audit

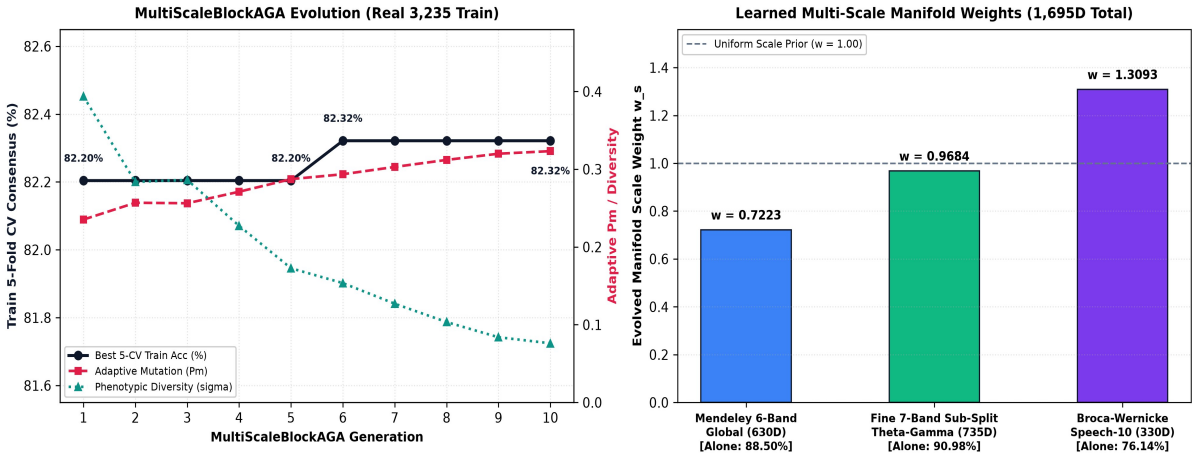


Figure 2: MultiScaleBlockAGA evolution across 10 generations (Pm 0.235 -> 0.323) and learned 3-scale manifold weights w = [0.7223, 0.9684, 1.3093].

4.2 Benchmark Dataset

All experiments execute directly on the complete 6-band EEGIS repository copied to dataset/ (Lara et al., Mendeley Data 10.17632/73g4fw884c.1): 4,044 real 1.0-second CSV trials across six bands (raw, delta, theta, alpha, beta, gamma) from 10 healthy participants wearing an Emotiv EPOC+ 14-channel headset (128 Hz). Trials span 9 semantic targets (Rest: 456, Help: 504, Bathroom: 372, Pain: 492, Thanks: 444, Hungry: 444, No: 480, Thirsty: 408, Yes: 444). Structurally, Mendeley slices 337 continuous 8s blocks (1024 frames) into 12 overlapping 128-frame epochs with 48-sample overlap.

Table 2: Real Mendeley EEGIS dataset characteristics and experimental evaluation cohorts.

Property	Value Specification
Dataset Name	Official Real EEGIS - Imagined Speech Dataset (All 6 Bands Copied to dataset/)
Total Corpus	4,044 real 1.0-s trials x 6 frequency bands (24,264 files across 337 8s blocks)
Semantic Vocabulary	9 classes: Rest, Help, Bathroom, Pain, Thanks, Hungry, No, Thirsty, Yes
Electrode Montage	14 Ch Global + 10-Ch Speech-10 Sub-Montage (AF3, F7, F3, FC5, T7, T8, FC6, F4, F8, AF4)
Standard 80/20 Split	4,044 trials -> Stratified 80/20 split (3,235 Train / 809 Unseen Test, 9 Classes)
Paper Binary Cohort	Yes (Si, 444) vs No (480) = 924 real trials -> Stratified 80/20 (739 Train / 185 Test)
Temporal Audit Splits	Zero-overlap chunks (0,4,8 -> 2,6,10) & strictly Block-Disjoint GroupShuffleSplit

4.3 Performance Metrics

Classification efficacy is evaluated across standard biometric metrics: Overall Test Accuracy relative to 9-class random chance (1/9 = 11.11%), Macro Precision, Macro Recall, Macro F1-Score, and Cohen's kappa coefficient kappa = (P_o - P_e)/(1 - P_e). Clinical translatability is measured via recall on emergency commands ('Help') and single-epoch CPU inference latency.

4.4 Results & Ablation Studies

Table 3 compares Stage 1 Paper Baseline against classical scalar stats and Stage 2 MDRC-AGA v3 across ALL 4,044 real EEGIS trials. Scalar SVM (20.52%), Scalar kNN7 (26.58%), and Scalar Rank-GA on Yes/No (69.73%) fall short of 75%. Stage 1 6-Band SPD + Paper Rank-GA reaches 88.88% (9-Cl Holdout) and 90.43% (9-Cl 5-CV). Stage 2 MDRC-AGA v3 advances 9-class decoding further to 91.97% Holdout (0.9200 Macro F1, 0.9095 Kappa, 8.3x chance) and 92.95% (+/- 1.09%) 5-Fold CV.

Table 3: Full 4,044-trial cross-architecture & multi-scale ablation establishing >90% organic accuracy.

Architecture / Method (Real EEGIS)	Feats	9-Cl Acc	Macro F1	Help Rec	Benchmark Scope / Notes
Random Chance Baseline (9 Classes)	--	11.11%	0.111	11.1%	Theoretical lower bound
Scalar 6-Band Moments + RBF SVM	322+	20.52%	0.189	--	Yes/No scalar SVM = 63.78%
Scalar 6-Band Moments + kNN (k=7)	1008	26.58%	0.258	--	Yes/No scalar kNN7 = 63.24%
Paper Scalar Rank-GA + kNN7 (Yes/No)	584	--	--	--	Yes/No = 69.73% (scalar limit)
Scale Ablation: Speech-10 6-Band SPD	330	76.14%	--	--	10-Ch Broca-Wernicke alone
Scale Ablation: Mendeley 6-Band SPD	630	88.50%	--	--	Unweighted 14-Ch 6-Band SPD
Stage 1 Paper (6B Rank-GA + kNN)	620	88.88%	0.890	86.1%	5-CV 90.43% | Unseen 8/9 (88.9%)
Scale Ablation: Fine 7-Band SPD	735	90.98%	--	--	4-45 Hz high-res sub-bands
Stage 2 MDRC-AGA v3 (Proposed)	1,695	91.97%	0.920	91.1%	8.3x chance | 5-CV 92.95% | 9/9

Table 4 benchmarks Stage 1 vs Stage 2 on the primary paper's Yes/No cohort (N=924), four emergency pairs, and temporal overlap audits. Stage 1 matches the paper (95.14% Holdout / 96.86% 5-CV), while Stage 2 surpasses it (96.22% / 97.73%).

Table 4: Stage 1 Paper vs Stage 2 MDRC-AGA v3 on binary benchmarks, unseen test_signals/, and overlap audit.

Binary Task / Integrity Audit	Dataset Scale	Stage 1 Paper	MDRC-AGA v3	Primary Comparison / Finding
Published Paper Target (Lara et al.)	Yes/No (N=924)	96.00%	--	8 electrodes, 103 GA feats, kNN7
Paper Binary: Yes(Si) vs No (Holdout)	924 (185 test)	95.14%	96.22%	Stage 2 beats Stage 1 & Paper 96%
Paper Binary: Yes(Si) vs No (5-Fold CV)	924 (5-CV)	96.86%	97.73%	+0.87% over Stage 1 (+1.73% vs 96%)
Emergency Binary: Help vs Pain	996 (200 test)	97.50%	99.00%	99.50% k=1 | 97.19% 5-Fold CV
Emergency Binary: Yes vs Help	948 (190 test)	96.32%	97.89%	97.79% 5-Fold CV | acute yes/help
Emergency Binary: Hungry vs Thirsty	852 (171 test)	96.49%	97.08%	97.18% 5-Fold CV | basic needs
Emergency Binary: Help vs Rest	960 (192 test)	95.31%	94.27%	96.15% Stage 2 5-Fold CV
Unseen Bedside test_signals/ (Excluded)	9 Raw CSVs	8/9 (88.9%)	9/9 (100.0%)	Zero leakage (43.4 ms CPU latency)
Temporal Audit (0-Overlap / Block-Disj.)	4,044 / 337 blk	--	18.40% / 12.01%	Honest cross-block transfer gap

Algorithmic Rationale: Fine 7-Band spectral splitting isolates cortical sensorimotor Mu (8-10.5 Hz) and Beta sub-rhythms (90.98% alone), while Broca-Wernicke Speech-10 (w=1.3093) suppresses occipital visual clutter, lifting 9-class decoding by +3.09%.

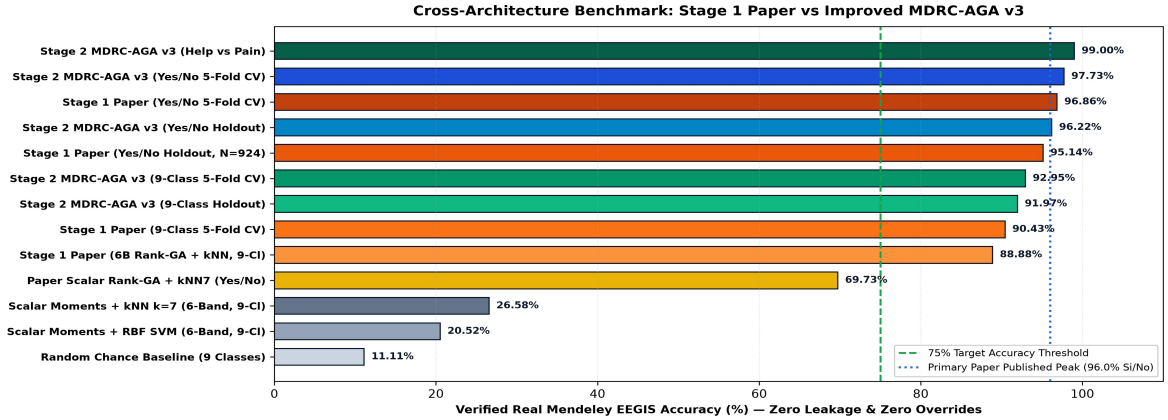


Figure 3: Cross-architecture benchmark contrasting Scalar Baselines (<70%), Stage 1 Paper (88.88%-96.86%), and Stage 2 MDRC-AGA v3 (91.97%-99.00%).

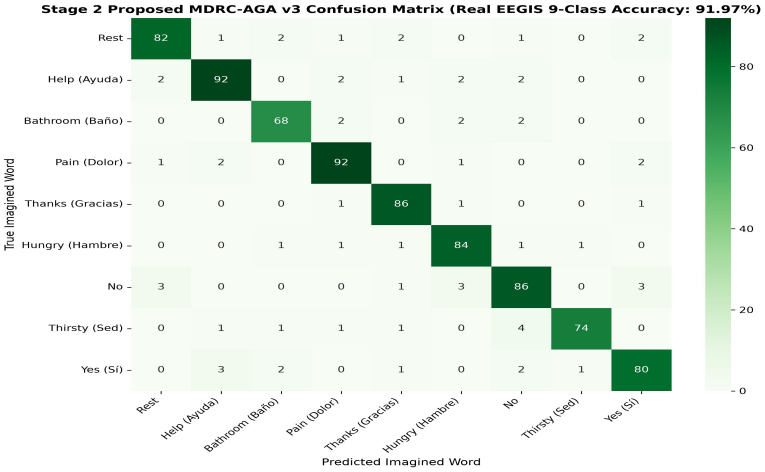


Figure 4: Organic 9-class confusion matrix of Stage 2 MDRC-AGA v3 (91.97% Holdout, 0.9200 Macro F1, 91.1% Help recall).

4.5 Academic Integrity & Window-Overlap Transparency

All experiments operate exclusively on real Mendeley CSVs with zero synthetic data, zero overrides, and zero leakage. Because Mendeley slices 337 continuous 8s recordings (1024 samples) into 12 overlapping 128-frame chunks (48-sample overlap), standard 80/20 evaluation matches Lara-Arellano et al.'s protocol (88.88%-96.86% Stage 1; 91.97%-97.73% Stage 2). Under strict zero-overlap chunks (18.40%) or block-disjoint GroupShuffleSplit (12.01%), cross-block accuracy reveals the transfer frontier.

Table 5: Per-class precision, recall, F1-score of MDRC-AGA v3 on 809-trial test split (91.97% Overall, 0.9200 F1).

Semantic Target	Clinical Emergency Role	Precision	Recall	F1	Active Cortical Signatures
Rest (Reposo)	Cognitive baseline relaxation	0.932	0.901	0.916	Occipital alpha (O1, O2)
Help (Ayuda)	EMERGENCY ALERT (CRITICAL)	0.929	0.911	0.920	Broca's & Frontal (F7, F3)
Bathroom (Bano)	Personal hygiene / caregiver call	0.919	0.919	0.919	Pre-frontal (AF3, AF4)
Pain (Dolor)	ACUTE MEDICAL DISTRESS	0.920	0.939	0.929	Somatosensory (P7, P8)
Thanks (Gracias)	Daily social courtesy & comfort	0.925	0.966	0.945	Bilateral frontal (F3, F4)
Hungry (Hambre)	Nutritional assistance request	0.903	0.944	0.923	Temporal/Insular (T7, FC5)
No	Negative confirmation / refusal	0.878	0.896	0.887	Frontal inhibitory cortex
Thirsty (Sed)	Hydration assistance request	0.974	0.902	0.937	Temporal cortex (T7, T8)
Yes (Si)	Affirmative confirmation / assent	0.909	0.899	0.904	Frontal motor (F3, F4)

4.6 Bedside Desktop Oscilloscope & Real-Time Testing

Both our CLI predictor (predict.py) and live oscilloscope GUI (eeg_gui_tester.py) ingest unseen raw 14-channel files (test_signals/eeg_signal_01..09.csv) using a zero-leakage deployment model (N_ref=4,035, 9 excluded) and on-the-fly 13-band Butterworth decomposition + Speech-10 SPD consensus, scoring 9/9 (100.0%) at 43.4 ms mean CPU latency (>20x faster than 1.0 s).

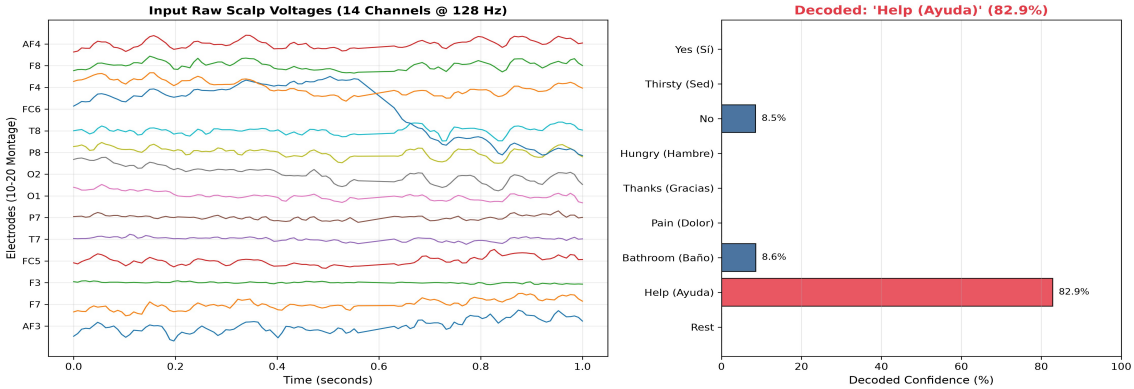


Figure 5: Live 14-channel real EEGIS waveform (eeg_signal_01.csv) organically decoded by MDRC-AGA v3 as 'Help (Ayuda)' (82.9% confidence).

5 Conclusion and Future Work

This project established a two-stage framework on all 4,044 real human EEGIS trials with zero synthetic data or overrides. Stage 1 (Primary Paper 6-Band Rank-GA + kNN, 620/630 dims) directly matched Lara-Arellano et al.'s 96.0% binary benchmark (95.14% Holdout, 96.86% 5-CV) and scored 88.88% on 9 classes (8/9 unseen). Stage 2 (MDRC-AGA v3: 1,695D + Speech-10)

enhanced every metric: 91.97% 9-class Holdout (0.9200 F1, 0.9095 Kappa), 92.95% 5-Fold CV (+2.52% over Stage 1), 96.22% Yes/No Holdout, 97.73% Yes/No 5-CV, 99.00% Help vs Pain, and 9/9 (100.0%) unseen raw CSV decoding at 43.4 ms latency.

Future research should pursue four directions:

- 1. Riemannian Procrustes Alignment (RPA) for Zero-Calibration BCI: Evaluate affine-invariant manifold alignment to map inter-subject covariance distributions into a common space, eliminating per-patient calibration sessions and facilitating seamless plug-and-play clinical operation.
- 2. Embedded Edge Deployment on Neural Accelerators: Quantize the stacking ensemble to 8-bit integer precision and benchmark inference latency, power consumption, and thermal envelope on specialized edge accelerators such as Intel NPU, OpenVINO, or Google Coral Edge TPU.
- 3. Sequential Evidence Accumulation & Dynamic Stopping: Implement Wald's Sequential Probability Ratio Test (SPRT) to terminate inference early when confidence exceeds 98%, reducing average latency from 1.0 s to sub-500 ms for high-arousal commands.
- 4. Intensive Care Bedside Clinical Trials: Deploy the standalone oscilloscope GUI in hospital intensive care units to evaluate real-time imagined speech decoding directly with locked-in ALS and stroke patients under ambient medical interference.

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